

# PANDUAN PEMBUATAN SMART GREENHOUSE IoT

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**Step-by-Step dari Nol sampai Jadi**

**ESP32-S3-N16R8 Soldered**

Panduan ini dirancang untuk orang awam.

Ikuti step-by-step dari atas ke bawah.

Jangan loncat step — setiap step bergantung pada step sebelumnya.

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## **FASE 0: Persiapan Software**

Sebelum menyentuh hardware, siapkan software dulu.

### **Step 1: Download & Install Arduino IDE**

1. Buka browser, pergi ke [arduino.cc/en/software](https://arduino.cc/en/software)
2. Download Arduino IDE 2.x (pilih sesuai OS: Windows/Mac/Linux)
3. Install seperti biasa (Next > Next > Finish)
4. Buka Arduino IDE setelah install selesai

### **Step 2: Install ESP32 Board Support**

Arduino IDE default hanya kenal Arduino Uno. Kita perlu menambahkan ESP32.

1. Buka Arduino IDE
2. File > Preferences
3. Di "Additional Board Manager URLs", paste:

**[https://espressif.github.io/arduino-esp32/package\\_esp32\\_index.json](https://espressif.github.io/arduino-esp32/package_esp32_index.json)**

4. OK > Tools > Board > Board Manager
5. Cari "esp32" > "esp32 by Espressif Systems" > Install
6. Tunggu selesai (5-10 menit)

### **Step 3: Install Library yang Dibutuhkan**

Library = kode tambahan supaya ESP32 bisa berkomunikasi dengan sensor.



Tools > Manage Libraries, cari & install:

- a. "BH1750" by Christopher Laws
- b. "DHT sensor library" by Adafruit
- c. "Adafruit Unified Sensor" by Adafruit
- d. "ESP32Servo" by Kevin Harrington
- e. "ArduinoJson" by Benoit Blanchon

## FASE 1: Rakit Hardware

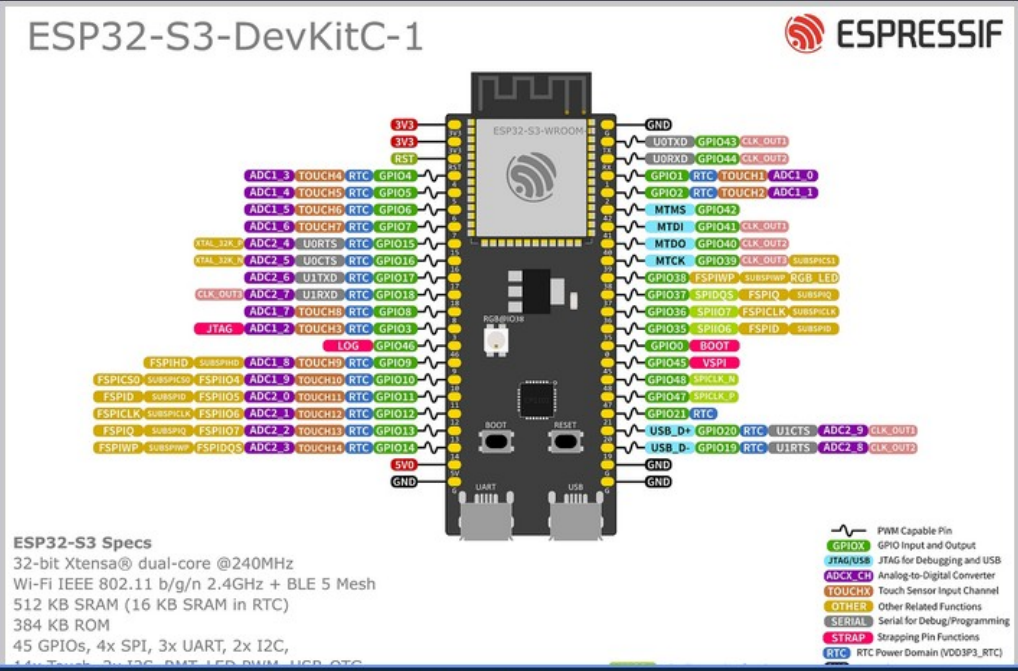
Rakit rangkaian elektronik di breadboard.

**PERINGATAN:** Jangan colok adaptor/USB sampai semua wiring selesai dan dicek!

### Step 4: Pasang ESP32 di Breadboard

Pasang ESP32-S3 di Breadboard 830 Point

Tancapkan di tengah di kedua sisi pin harus bisa diakses



**ESP32-S3 Specs**  
32-bit Xtensa® dual-core @240MHz  
Wi-Fi IEEE 802.11 b/g/n 2.4GHz + BLE 5 Mesh  
512 KB SRAM (16 KB SRAM in RTC)  
384 KB ROM  
45 GPIOs, 4x SPI, 3x UART, 2x I2C,  
14x Touch, 3x I2C, PWM, LED PWM, MCP, RTC

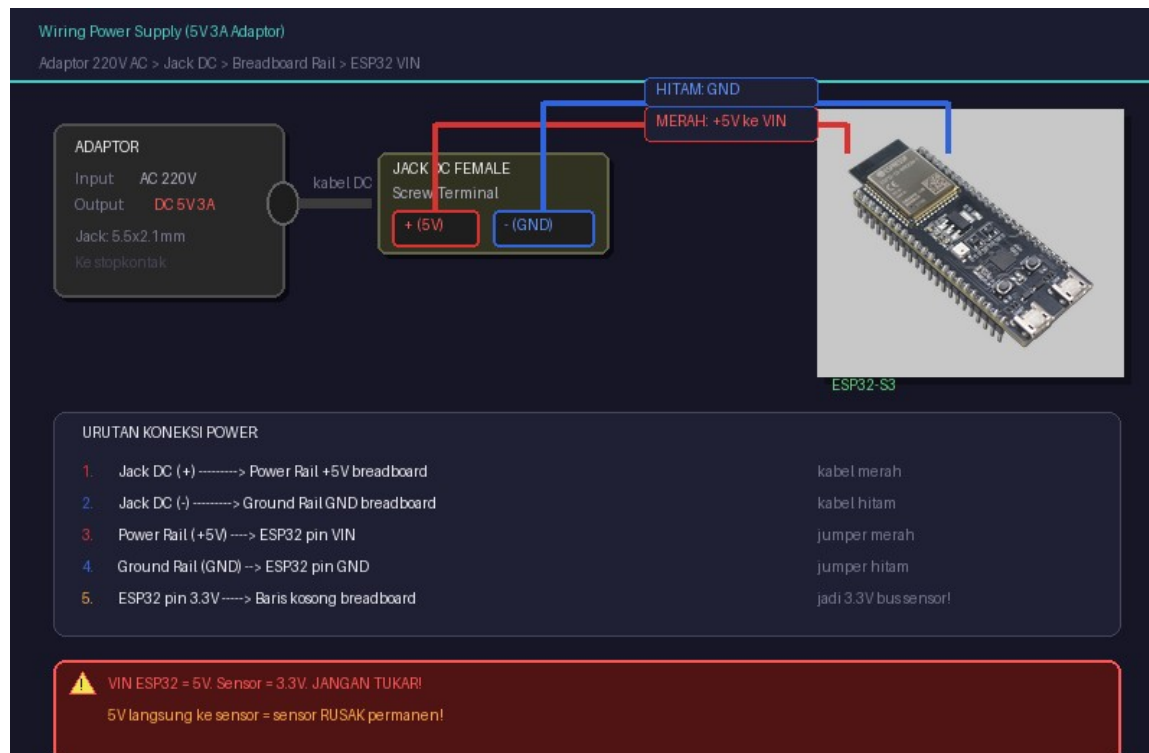
Tekan ESP32 kuat semua pin masuk lubang. Kedua sisi harus bisa dipasangi jumper.

**Legend:**  
PWM Capable Pin  
GPIOX GPIO Input and Output  
JTAG/USB JTAG for Debugging and USB  
ADCX CA Analog-to-Digital Converter  
TOUCHX Touch Sensor Input Channel  
OTHERX Other Related Functions  
SERIALX Serial for Debug/Programming  
STRAPX Strapping Pin Functions  
RTCX RTC Power Domain (VDD3P3\_RTC)

1. Ambil breadboard 830 point
2. Tancapkan ESP32-S3 di TENGAH breadboard
3. Kedua sisi pin harus bisa diakses untuk jumper wire

**TIP:** Tekan kuat tapi hati-hati. Semua pin harus masuk.

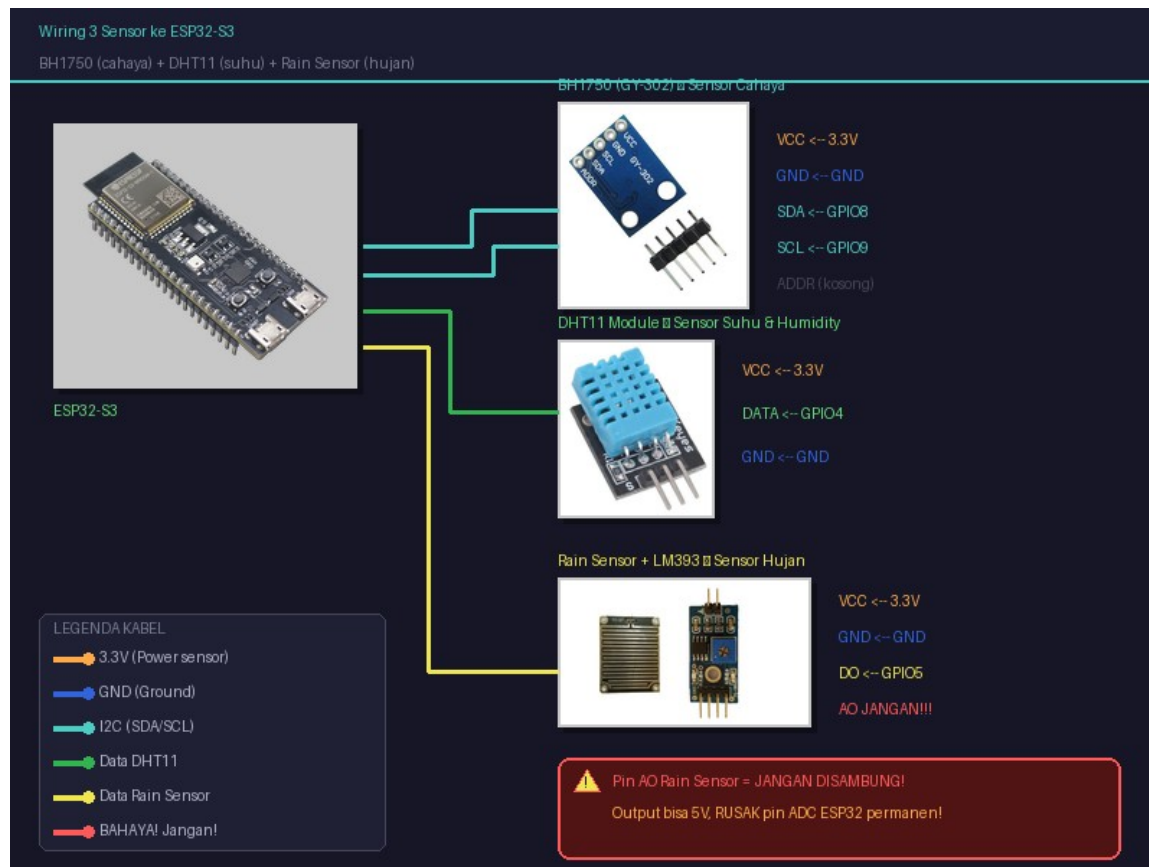
## Step 5: Sambung Power Supply



1. Adaptor 5V 3A > jack DC female screw terminal (belum colok!)
2. Terminal (+) > power rail (+) breadboard = +5V
3. Terminal (-) > ground rail (-) breadboard = GND
4. Power rail (+) > ESP32 pin VIN
5. Ground rail (-) > ESP32 pin GND
6. ESP32 pin 3.3V > baris baru breadboard (jadi 3.3V bus sensor)

**PERINGATAN: VIN ESP32 = 5V. Sensor = 3.3V. Jangan tukar!**

## Step 6: Sambung Sensor Cahaya (BH1750)



BH1750 pakai I2C (SDA + SCL):

VCC --> 3.3V bus

GND --> GND bus

SDA --> GPIO8

SCL --> GPIO9

ADDR --> kosong

## Step 7: Sambung Sensor Suhu (DHT11)

DHT11 module 3-pin:

VCC --> 3.3V bus

GND --> GND bus

DATA --> GPIO4

TIP: Beli yang MODULE 3-pin (ada PCB), bukan komponen telanjang 4-pin.

## Step 8: Sambung Sensor Hujan (Rain Sensor)

Dari control board LM393:

VCC --> 3.3V bus

GND --> GND bus

DO --> GPIO5

AO --> JANGAN DISAMBUNG!

**PERINGATAN:** Pin AO bisa output 5V. ESP32 ADC max 3.3V. Disambung = RUSAK PERMANEN.

## Step 9: Sambung Servo Motor (MG996R)

Wiring Servo MG996R + Kapasitor 470uF  
POWER SERVO = LANGSUNG DARI ADAPTOR 5V!



Servo MG996R  
Torque: 11 kg.cm | 4.8-7.2V

Pinout MG996R (3 kabel warna)

|                       |                                     |
|-----------------------|-------------------------------------|
| Kabel MERAH (VCC)     | +5V Power Rail (ADAPTOR!)           |
|                       | Langsung dari adaptor, BUKAN ESP32! |
| Kabel COKLAT (GND)    | GND Rail (ground bus)               |
|                       | Ground bersama semua komponen       |
| Kabel ORANYE (Signal) | GPIO13 ESP32                        |
|                       | Sinyal PWM, kontrol posisi servo    |

⚠️ SERVO POWER = ADAPTOR 5V! BUKAN dari pin ESP32!  
ESP32 max 500mA. Servo stall = 2500mA. Bisa RUSAK ESP32!

MERAH (VCC) --> +5V Rail (ADAPTOR!)

COKLAT (GND) --> GND bus

ORANYE (Signal) --> GPIO13

Capacitor 470uF dekat servo:

Kaki panjang (+) --> +5V Rail



Kaki pendek (-) --> GND Rail

**PERINGATAN: Servo power WAJIB dari adaptor. BUKAN ESP32! Servo butuh 2.5A, ESP32 max 500mA.**

## Step 10: Sambung LED Warning + Resistor

Wiring LED Warning + Resistor 220 Ohm  
GPIO2 --> Resistor 220 Ohm --> LED Merah --> GND



LED Merah 5mm  
Kaki panjang = + (Anoda)  
Kaki pendek = - (Katoda)

RANGKAIAN SERI:

```
graph LR; ESP32[ESP32 GPIO2] -- arus --> R[RESISTOR 220 Ohm]; R --> LED((LED Merah)); LED --> GND[GND];
```

DETAIL:

|                   |                                                             |
|-------------------|-------------------------------------------------------------|
| Resistor 220 Ohm: | Tidak ada polaritas (boleh dibalik)                         |
| LED 5mm:          | Kaki PANJANG = Anoda (+), Kaki PENDEK = Katoda (-)          |
| Fungsi:           | Berkedip saat Anti-Oven warning aktif (tutup + panas > 33C) |
| Kenapa 220 Ohm?   | Membatasi arus ~15mA supaya LED tidak terbakar              |

GPIO2 > Resistor 220 Ohm > LED Merah > GND

Resistor: tidak ada polaritas. LED: kaki panjang = (+).

## FASE 2: Upload & Test Firmware

### Step 11: Checklist Sebelum Power On

**CHECKLIST SEBELUM POWER ON**  
Cek SEMUA sebelum colok adaptor ke listrik!

☒

**WAJIB** Semua GND terhubung ke SATU ground bus

☒

**WAJIB** Tidak ada kabel 5V masuk langsung ke sensor (harus 3.3V!)

☒

**WAJIB** Rain sensor pin AO TIDAK tersambung ke manapun

☒

**WAJIB** Servo power dari ADAPTOR 5V, BUKAN dari pin ESP32

☒

**WAJIB** Capacitor 470uF polaritas benar: (+) ke 5V, (-) ke GND

☒

**WAJIB** Tidak ada kabel/komponen yang saling menyentuh (short)

☒

ESP32 belum di-power (belum colok adaptor/USB)

☒

Kabel USB yang disiapkan = kabel DATA (bukan charge-only)



Kalau ada yang SALAH = komponen RUSAK PERMANEN!  
Double-check! Lebih baik lambat tapi aman.

**Cek SEMUA item sebelum colok adaptor!**

Setelah yakin:

1. Colok adaptor > cek LED power ESP32 menyala
2. Tunggu 10 detik > sentuh komponen
3. Kalau ada yang PANAS > CABUT SEGERA

## Step 12: Colok ESP32 ke Laptop via USB

Colok ESP32-S3 ke Laptop via USB-C

Koneksi pertama: ESP32 ke laptop dengan kabel USB-C DATA



Device Manager

Ports (COM & LPT)

COM3 as USB Serial  
(ESP32-S3 detected)

Kalau ini MUNCUL =  
kabel USB BENAR

KABEL  
USB-C  
DATA  
USB-A

ESP32-S3-N16R8 Soldered

⚠ Kabel HARUS USB DATA!  
Kabel charge-only TIDAK punya jalur data D+/D-

Cara cek: Device Manager  
Harus muncul COM port baru (COM3/COM4)

Data cable: 4 kabel internal (VCC, D+, D-, GND) bisa transfer data  
Charge-only: 2 kabel internal (VCC, GND saja) TIDAK bisa transfer data

1. Kabel USB-C DATA (bukan charge-only!)
2. Colok ke ESP32 dan laptop
3. Device Manager harus muncul COM port baru

Tidak muncul? Ganti kabel USB atau install driver CH340/CP2102.

## Step 13: Konfigurasi Arduino IDE

Setting Board di Arduino IDE

Tools > Board Settings setiap setting yang merah = KRITIS

Arduino IDE 2.x > Tools > Board Configuration

Board: ESP32S3 Dev Module  
Board manager ESP32 by Espressif

PSRAM: OPI PSRAM  
KRITIS! Harus OPI, BUKAN QSPI!

Flash Size: 16MB (128Mb)  
Sesuai spek board N16R8

USB CDC On Boot: Enabled  
KRITIS! Tanpa ini Serial Monitor MATI

Upload Speed: 921600  
Speed tercepat

Partition Scheme: Default 4MB with spiffs  
Default cukup

Port: COM3 (atau yg muncul)

Board yang dipakai:  
ESP32-S3-N16R8

⚠ PSRAM = OPI & USB CDC = Enabled WAJIB BENAR!  
Salah setting: board tidak terdeteksi atau PSRAM out of memory

**Board: ESP32S3 Dev Module**

**PSRAM: OPI PSRAM**

**Flash Size: 16MB (128Mb)**

**USB CDC On Boot: Enabled**

**Upload Speed: 921600**

**Port: COM yang muncul**

**PERINGATAN: PSRAM = OPI & USB CDC = Enabled WAJIB BENAR!**

### **Step 14: Siapkan File Firmware**

1. Clone/download folder firmware/ dari repo
2. Buka smart\_greenhouse.ino di Arduino IDE
3. Copy credentials.h.example > credentials.h
4. Edit credentials.h:

```
#define WIFI_SSID "nama_wifi"
```

```
#define WIFI_PASSWORD "password_wifi"
```

```
#define SUPABASE_URL "https://xxx.supabase.co"
```

```
#define SUPABASE_KEY "eyJ..."
```

TIP: Isi WiFi dulu. Supabase nanti. Sistem tetap jalan offline.

## Step 15: Upload Firmware ke ESP32

Upload Firmware ke ESP32-S3

Klik Upload > Compile > Upload > Done!

- 1 Buka smart\_greenhouse.ino  
File > Open di Arduino IDE
- 2 Board = ESP32S3 Dev Module  
Tools > Board
- 3 Port = COM yang muncul  
Tools > Port
- 4 Klik **UPLOAD**  
Ikon panah kanan (Ctrl+U)
- 5 'Compiling sketch...'  
Tunggu 1-2 menit
- 6 'Uploading...'  
Progress bar berjalan
- 7 **SELESAI!**  
'Hard resetting via RTS pin...'

**TROUBLESHOOT: Gagal 'Connecting.....'**

1. Tahan tombol BOOT di board ESP32
2. Sambil tahan BOOT, tekan RESET sekali
3. Lepas RESET (BOOT masih ditahan)
4. Klik Upload di Arduino IDE
5. Setelah 'Uploading...' muncul, lepas BOOT

Klik Upload > tunggu compile > tunggu upload.

Kalau gagal "Connecting...": tahan BOOT > tekan RESET > lepas > upload > lepas BOOT

## Step 16: Buka Serial Monitor

```
Serial Monitor #Contoh Output Normal
Tools > Serial Monitor | baud rate = 115200

Serial Monitor #COM3 #115200 baud

=====
SMARTGREENHOUSE IoT
Board: ESP32-S3-N16R8 Soldered
Firmware: All Phases Active
=====
Free heap : 287456 bytes
Free PSRAM : 8388608 bytes
=====

[WIFI] Menghubungkan ke MyWiFi...
[WIFI] Terhubung! IP: 192.168.1.47
[WEATHER] Prediksi 6 jam: hujan=TIDAK (25%)
[SYSTEM] Inisialisasi selesai.

-----
Suhu   : 28.5 C
Humidity : 65.2 %
Cahaya : 15420.0 lux
Hujan  : TIDAK
Atap    : BUKA (180 derajat)
Mode    : AUTO
Overheat : Normal

Suhu/Humidity = -1.0 -> DHT11 error!          Cahaya = -1.0 -> BH1750 error!
Cek kabel sensor yang error tersebut.
```

Tools > Serial Monitor, baud rate 115200.

Tekan RESET di ESP32. Harus muncul data sensor.

Kalau ada -1 = sensor error, cek kabelnya.

## FASE 3: Setup Supabase

### Step 17: Buat Akun & Project Supabase



supabase.com > Sign Up > New Project > smart-greenhouse

Region: Southeast Asia (Singapore)

### Step 18: Catat URL dan API Key

Settings > API:

**Project URL:** https://xxxxx.supabase.co

**anon public key:** eyJ...

Update credentials.h > upload ulang firmware.

**PERINGATAN:** Pakai "anon public" BUKAN "service\_role" key!

### Step 19: Buat Tabel Database

SQL Editor > New Query > paste & Run:

```
CREATE TABLE sensor_logs (  
  id BIGSERIAL PRIMARY KEY, temperature FLOAT, humidity FLOAT,  
  lux FLOAT, is_raining BOOLEAN DEFAULT FALSE,  
  roof_angle INTEGER DEFAULT 0, roof_state TEXT DEFAULT 'closed',  
  mode TEXT DEFAULT 'auto', overheating BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMPTZ DEFAULT NOW())
```

```
);  
CREATE TABLE commands (  
  id BIGSERIAL PRIMARY KEY, action TEXT NOT NULL,  
  executed BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMPTZ DEFAULT NOW(), executed_at TIMESTAMPTZ  
);  
ALTER TABLE sensor_logs ENABLE ROW LEVEL SECURITY;  
ALTER TABLE commands ENABLE ROW LEVEL SECURITY;  
CREATE POLICY "read" ON sensor_logs FOR SELECT USING (true);  
CREATE POLICY "insert" ON sensor_logs FOR INSERT WITH CHECK (true);  
CREATE POLICY "all" ON commands FOR ALL USING (true) WITH CHECK (true);
```

## Step 20: Verifikasi Data Masuk

Tunggu 60 detik > Table Editor > sensor\_logs > harus ada data baru!



## FASE 4: Test Sistem Lengkap

### Step 21: Test: Sensor Baca Normal

Serial Monitor tiap 5 detik: suhu (20-35C), humidity (40-90%), cahaya (0-65000 lux).

### Step 22: Test: Hujan = Atap Tutup (Rule 1)



Teteskan air ke rain plate > "Hujan: YA" > tunggu 60 detik > servo tutup (0 derajat).

Rule 1: Local is King. Tidak bisa di-override.

### Step 23: Test: Cerah = Atap Buka

Keringkan rain plate > arahkan senter ke BH1750 > tunggu 60 detik > servo buka (180).

### Step 24: Test: Hysteresis

Tetes air > tunggu 30 detik > keringkan > servo TIDAK BOLEH bergerak!

### Step 25: Test: Anti-Oven (Rule 4)

Hujan aktif + suhu > 33C: LED berkedip, atap tetap tutup.

## **FASE 5: Web Dashboard (Opsional)**

### **Step 26: Setup Next.js**

`npx create-next-app@latest greenhouse-dashboard`

`npm install @supabase/supabase-js`

`.env.local`: NEXT\_PUBLIC\_SUPABASE\_URL & NEXT\_PUBLIC\_SUPABASE\_ANON\_KEY

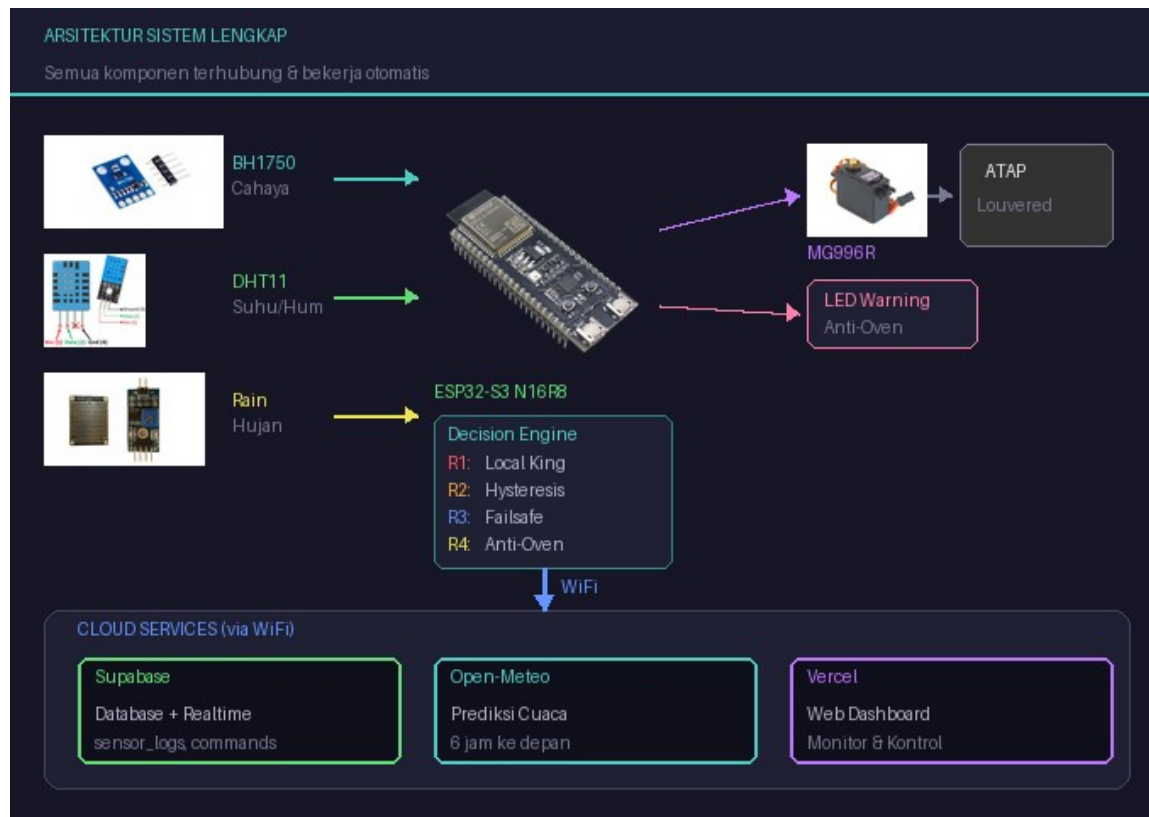
### **Step 27: Deploy ke Vercel**

Push ke GitHub > vercel.com > Import > Add env vars > Deploy

### **Step 28: Test Dashboard**

Data sensor muncul, tombol BUKA/TUTUP/AUTO berfungsi.

## SISTEM LENGKAP



**Selamat! Smart Greenhouse IoT sudah berjalan.**

Sensor tiap 5s | Supabase tiap 60s | Command tiap 10s | Weather tiap 30min | Offline OK